

# Parth Deshpande

+919770514488 | [deshpandeparth793@gmail.com](mailto:deshpandeparth793@gmail.com) | [linkedin.com/in/parth-deshpande](https://linkedin.com/in/parth-deshpande) | [github.com/Parth-bot-crypto26](https://github.com/Parth-bot-crypto26)

## Education

### VIT Bhopal University, Bhopal

June 2023 – June 2027

Bachelor of Technology in Computer Science & Engineering (AI&ML)

CGPA: 9.0/10

### Govt. Excellence H. S. School, Betul

2023

Class 12th Board: 94%

Class 10th Board: 100%

## Technical Skills

**Languages:** Python, C++, JavaScript (ES6+), SQL, Java

**AI & Machine Learning:** PyTorch, TensorFlow, Hugging Face, NLP, OpenCV, LLMs

**Web Development:** React.js, Node.js, Express, HTML5, CSS3, RESTful APIs

**Databases:** PostgreSQL, MongoDB, MySQL, FAISS (Vector DB), Firebase

**Developer Tools & Cloud:** Git, Docker, AWS (EC2, S3), MLflow, Postman, VS Code, Linux

## Projects

### Vector-Database: High-Dimensional Vector Search Engine

Jun 2026 – Jul 2026

*Technologies:* C++, HNSW, KD-Tree, cpp-httplib (REST API), Ollama, RAG

- Engineered a vector database from scratch in C++, implementing HNSW, KD-Tree, and Brute-Force search side-by-side across 3 configurable distance metrics (cosine, Euclidean, Manhattan), benchmarking  $O(\log N)$  approximate retrieval against exact  $O(N \cdot d)$  baselines.
- Built a full RAG pipeline integrating local LLM inference (Ollama `nomie-embed-text` + `llama3.2`) with the custom HNSW index, chunking documents into overlapping 250-word segments and streaming token-level answers over a REST API.
- Designed a real-time PCA-based 2D visualization dashboard to project 16D/768D embeddings and expose semantic clustering across demo vectors and user-uploaded documents.

### Semantic Code Similarity: NLP-Based Plagiarism Detector

Feb 2026 – Mar 2026

*Technologies:* Python, Word2Vec, GloVe, NLP, scikit-learn, Streamlit

- Built an NLP-based plagiarism detector that compares Python source code semantically rather than syntactically, using tokenization, identifier normalization, and mean-pooled Word2Vec/GloVe embeddings.
- Implemented dual embedding backends and configurable cosine-similarity thresholds to flag code obfuscated via variable renaming, comment edits, or reformatting; deployed an interactive Streamlit UI for two-snippet and multi-file batch analysis.
- Designed a pairwise similarity heatmap for folder-level plagiarism scans across an entire codebase in a single pass.

### Subway Surfers Virtual Play: Real-time HCI System

Dec 2025 – Jan 2026

*Technologies:* Python, OpenCV, MediaPipe, PyAutoGUI

- Engineered a real-time Human-Computer Interaction system controlling Subway Surfers through live body pose detection, eliminating keyboard input entirely via gesture-to-keypress mapping.
- Designed a calibration and landmark-tracking pipeline using MediaPipe Pose to classify 5 game actions from shoulder and hip landmark geometry.
- Structured the project into modular utility files with dedicated test scripts per gesture, enabling isolated debugging and rapid iteration.

## Achievements

- Selected for Amazon ML Summer School 2026 — top 2.2% (~3,000) of 1.34 lakh+ registrations
- Attained Pupil Rank (Max Rating: 1409) on Codeforces
- Mastered DSA — 900+ problems on LeetCode, Codeforces & CodeChef
- Finalist, HackTU 6.0 — top team among 3,500+ students, 800+ teams

## Certifications

- Certified Data Science Associate Program — ICTRD (Indian Council for Technical Research Development)
- Introduction to Agentic AI — Cognizant Skillspring
- Fundamentals of Generative AI — Cognizant Skillspring
- Oracle Agentic AI Certified Foundations Associate
- 5-Day AI Agents: Intensive Vibe Coding Course With Google — Kaggle
- Applied Machine Learning in Python — University of Michigan (Coursera)